



HORMONES, SEXUAL HEALTH & MISC

Thymosin Alpha-1

Thymosin Alpha-1 – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Thymalfasin; thymosin alpha-1
SYNONYMS / ALIASES	Talpha1; thymosin alpha 1; thymalfasin; ZADAXIN
CAS NUMBER	62304-98-7
MOLECULAR FORMULA	C129H215N33O55
MOLECULAR WEIGHT	3108.31 g/mol
AMINO ACID SEQUENCE	Ac-SDAAVDTSSSEITTKDLKEKKEVVEEAEN (28 residues; N-terminus acetylated; sequence corresponds to the native human thymosin alpha-1)
SOURCE	Synthetic (solid-phase peptide synthesis with N-terminal acetylation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Contains no cysteines or methionines; relatively oxidation-stable. Avoid repeated freeze-thaw cycles.

STRUCTURAL NOTES

Thymosin alpha-1 is a synthetic 28-amino-acid peptide whose sequence corresponds to the N-terminal 28 amino acids of prothymosin alpha, a thymic gene product. The peptide is N-acetylated at the alpha-amino terminus, matching the native post-translationally modified form. It contains a high proportion of acidic (Asp, Glu) residues, giving the peptide a strongly acidic isoelectric point and a net negative charge at physiological pH. There are no cysteines, methionines, or tryptophan in the sequence.

SELECTED PEER-REVIEWED REFERENCES

1. Goldstein A.L., Badamchian M. - *Thymosins: chemistry and biological properties in health and disease* - Expert Opinion on Biological Therapy, 2004 - DOI: 10.1517/14712598.4.4.559
2. Garaci E. et al. - *Thymosin alpha 1: from bench to bedside* - Annals of the New York Academy of Sciences, 2007 - DOI: 10.1196/annals.1415.022
3. King R., Tuthill C. - *Immune modulation with thymosin alpha 1 treatment* - Vitamins and Hormones, 2016 - DOI: 10.1016/bs.vh.2016.04.003

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.